



Towards Standardized Benchmarking of SGRT Technologies: From Commissioning to Long-Term Quality Assurance

Siegfried Kollotzek | T. Buck · N. Bersenkovitsch · L. Bodner · P. Entfellner · U. Ganswindt

Department of Radiation Oncology, Medical University of Innsbruck / Tirol Kliniken, Austria

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SGRT is everywhere — its QA is not

AAPM TG302: “As recommended by TG-147, these tests should be performed by a QMP.”

Al-Hallaq et al., Med. Phys. 2022;49:e82–e112

Austria: § 61 StrSchG (ÖNORM S 5290-1:2011)
→ Surface-Scanner QA rests with the medical physicist

ESTRO-ACROP Table 4 — QA tests, C-arm linac (excerpt, simplified)

ID	Test	Tolerance	Freq.
A1	Isocentre (laser / radiographic)	1 mm / 1°	D / M
A5	Couch rotation	1 mm / 1°	M / A
E1	End-to-end positioning test	2 mm / 1°	A / R
D1	Beam hold performance	2 % / 2 mm	A
D2	Tracking performance (transl. / rot.)	1 mm / 1°	M / A
D3	Respiratory trace	pass	D / M
P1	Thermal drift (clinical)	0.5 mm	A
S1	Interlocks	pass	M / A

Source: Freisleder et al., Radiother Oncol 2022, Table 4 (re-typeset;
D/M/A = daily/monthly/annually, R = after repair)

Guidelines define what to test, how often, and to what tolerance — but not how.

Fig. 1 ESTRO-ACROP Table 4, excerpt

Our approach: classify, then close the gap

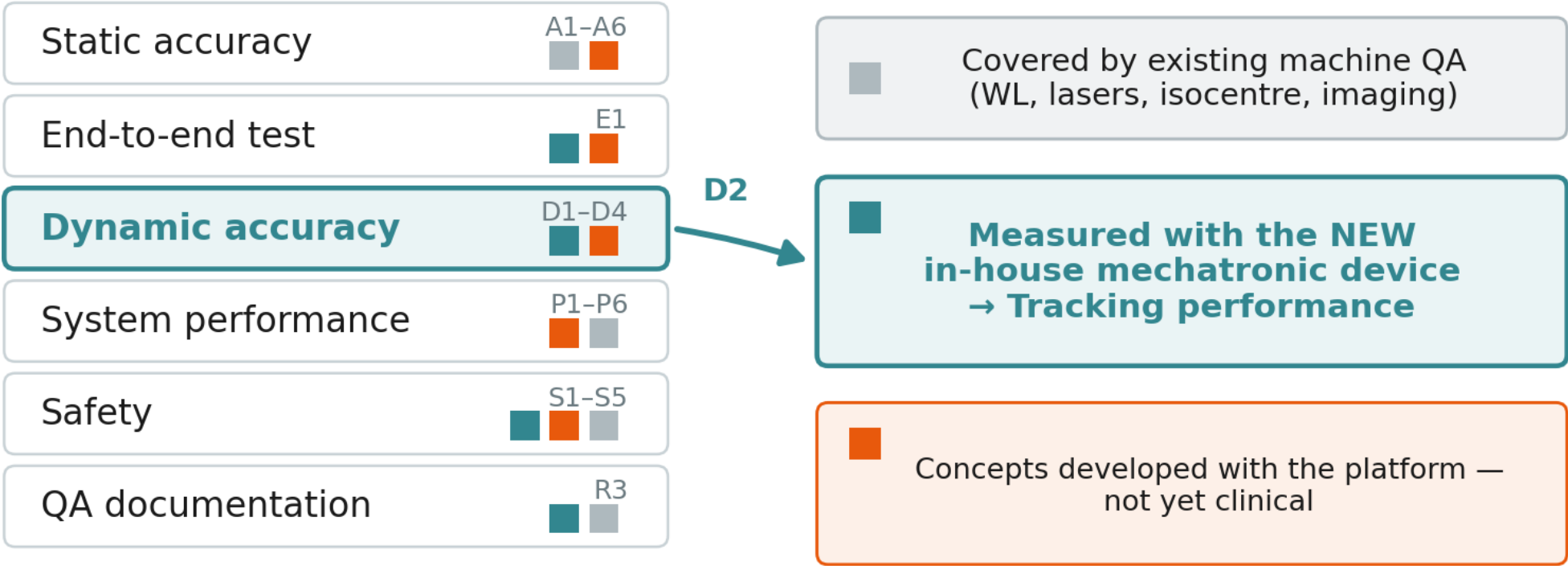
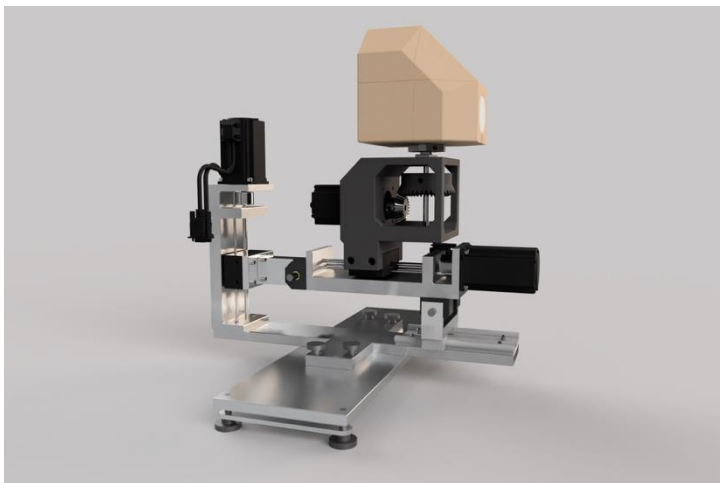


Fig. 2 Classification of the Table-4 parameters for C-arm linac environment

The tool: an open mechatronic motion platform

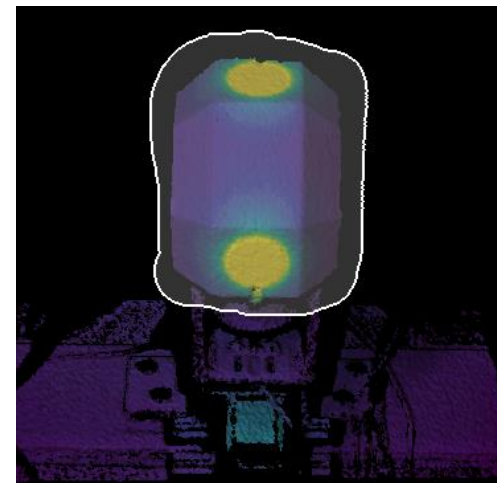
CAD animation



Clinical setup



Thermal signature (32 °C)



kV X-ray, embedded sphere



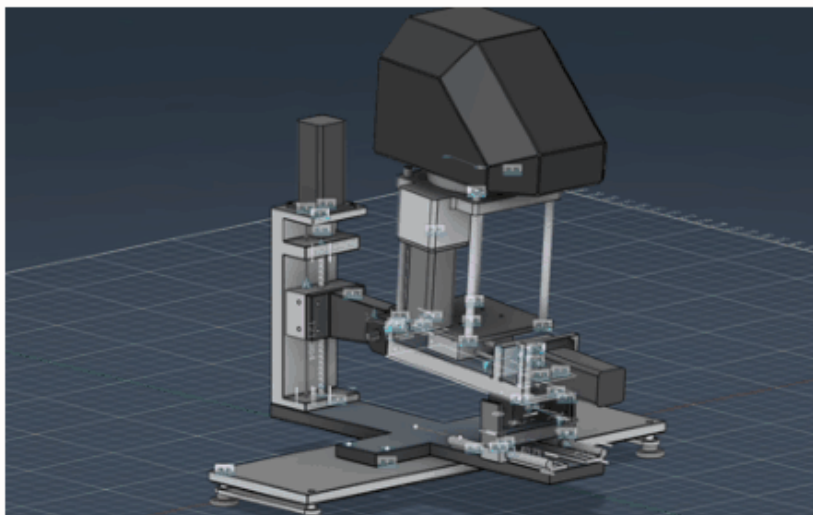
Fig. 4 The platform in four views: 3 closed-loop motors ($1.25 \mu\text{m}/\text{step}$) $\rightarrow$ 6 clinical DoF; heated surface; 8mm steel sphere as ground truth.

Blueprint-driven
Engineering &
analysis chain
Poster T. Buck



How the data is born

Platform — CAD animation



settling

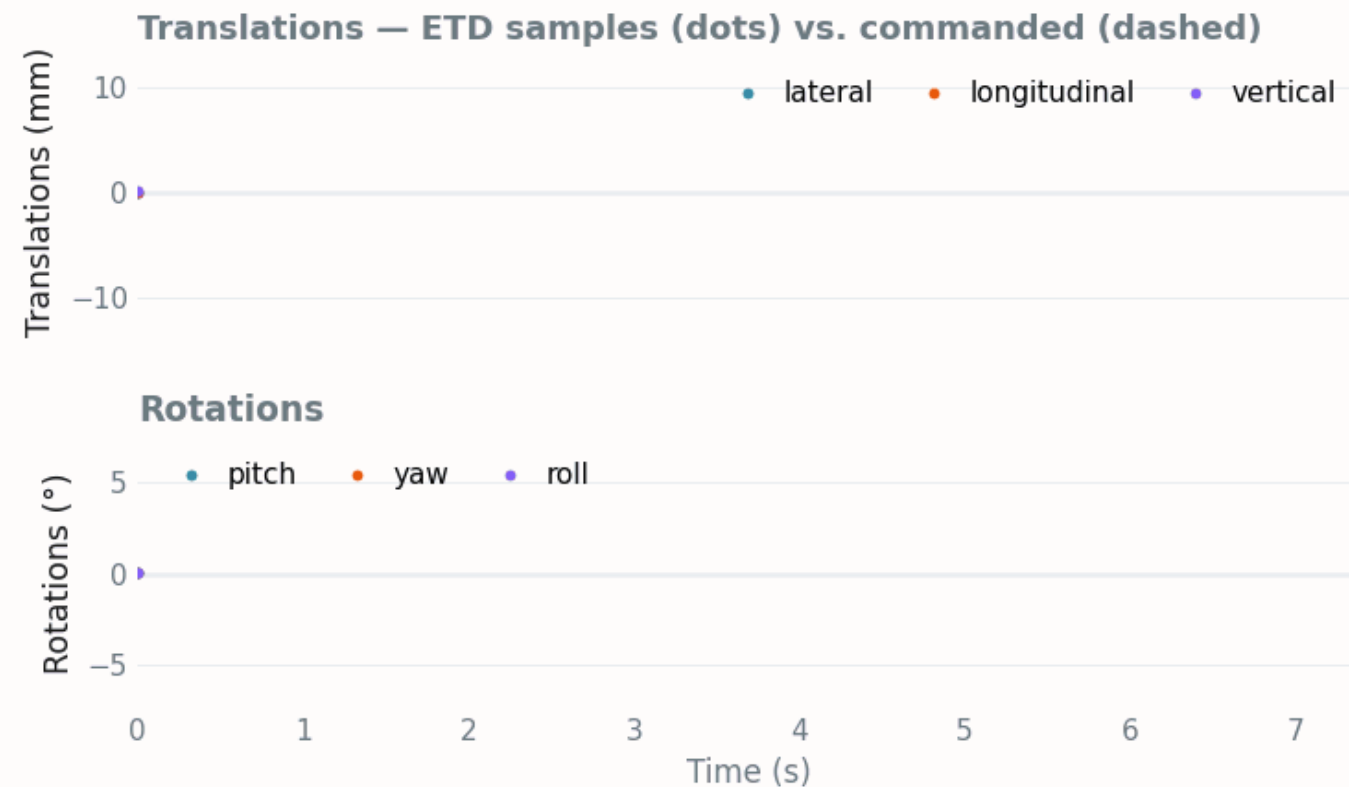


Fig. 5 All-axes sequence, drawn in sync with the motion: pure horizontal · pure vertical (→ vertical + pitch, coupled) · pure yaw

Trust the ground truth

< 0.1 mm

mechanical positioning reproducibility

0.0–0.1 mm

sphere-detection error on a
commanded 10 mm move

0.05–0.23 mm

RMSE after per-linac kinematic
calibration

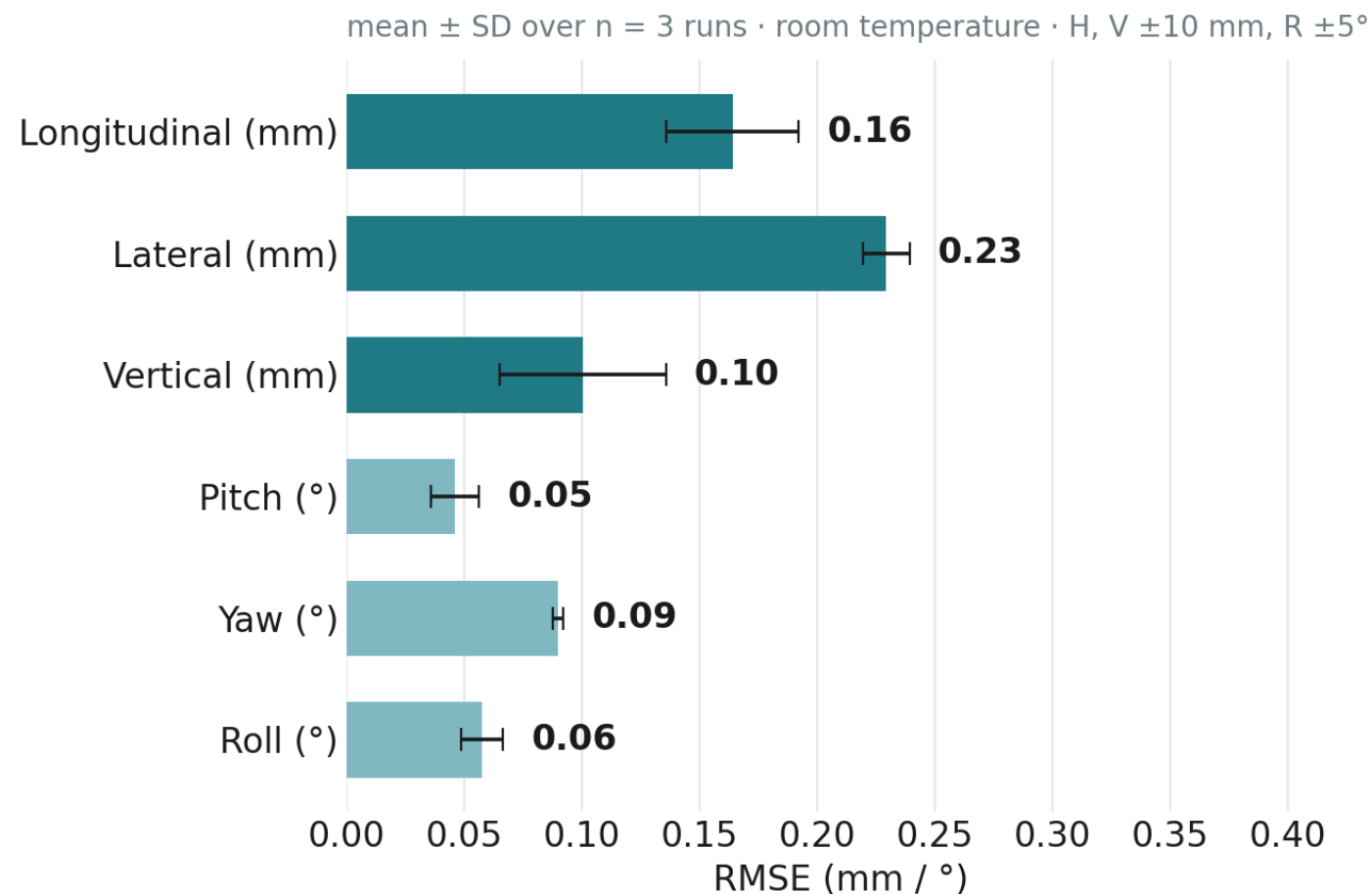


Fig. 6 All-axes validation sequence — RMSE per DoF (mean $\pm$ SD, $n = 3$, RT)

Characterisation: performance — and its limits

- 36 runs: isolated H/V/R · combined · variable speeds · RT / 32 °C
- Beyond the clinic: 90° rotations, speed ×100, yaw in 1° steps
- Two reproducible tracking-loss modes found

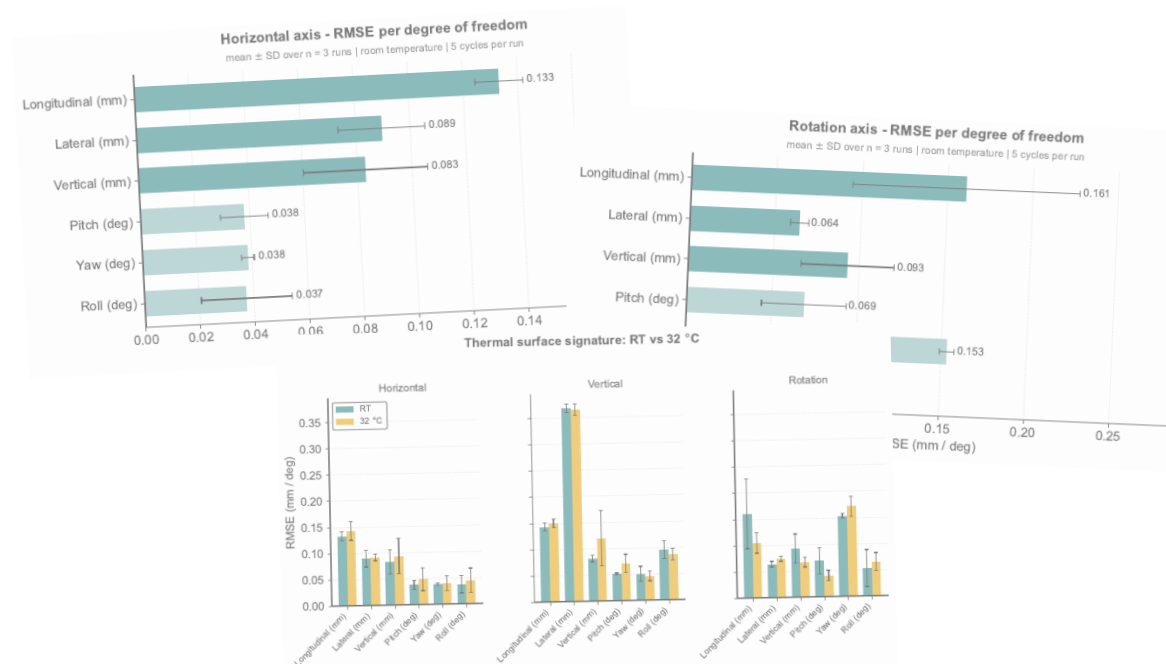


Fig. 7 Characterization preview — horizontal / rotation / thermal comparison

Knowing where it breaks is a QA result: those modes become interlock & robustness tests.

→ **full analysis & statistics: manuscript in preparation**

From characterisation to routine: the annual QA

- Deflection 1: everyday-scale
- Deflection 2: large but plausible
- 5 sphere detections each run
- Scripted chain:
 - raw logs
 - MAE / RMSE / MaxAE
 - Report

Passrates for all 6 DOF				
DOF	pass [%]	watch [%]	action [%]	results
longitudinal	100	0	0	Pass
lateral	100	0	0	Pass
vertical	100	0	0	Pass
roll	100	0	0	Pass
pitch	100	0	0	Pass
yaw	100	0	0	Pass

Maximum error metrics		
Max. MAE	Max. RMSE	Max. Absolute Error
0.215 mm	0.248 °	0.673 °

Applied tolerance levels and evaluation criteria			
Metric	Pass	Watch	Action
Tracking performance (translation)	≤ 1 mm	1 – 2 mm	> 2 mm
Tracking performance (rotation)	≤ 1°	1 – 2°	> 2°
Criterion			
Pass rate [%]	≥ 99	95–99	< 95
Watch rate [%]	≤ 1	1–5	–
Action rate [%]	0	0	> 0

Fig. 8 Auto-generated annual QA report (L1, 2026) — pass rates & maximum error

The annual QA in practice — one unit, four runs

ETDS QA Overview - L1 | Pads 32°C | Couch 0deg

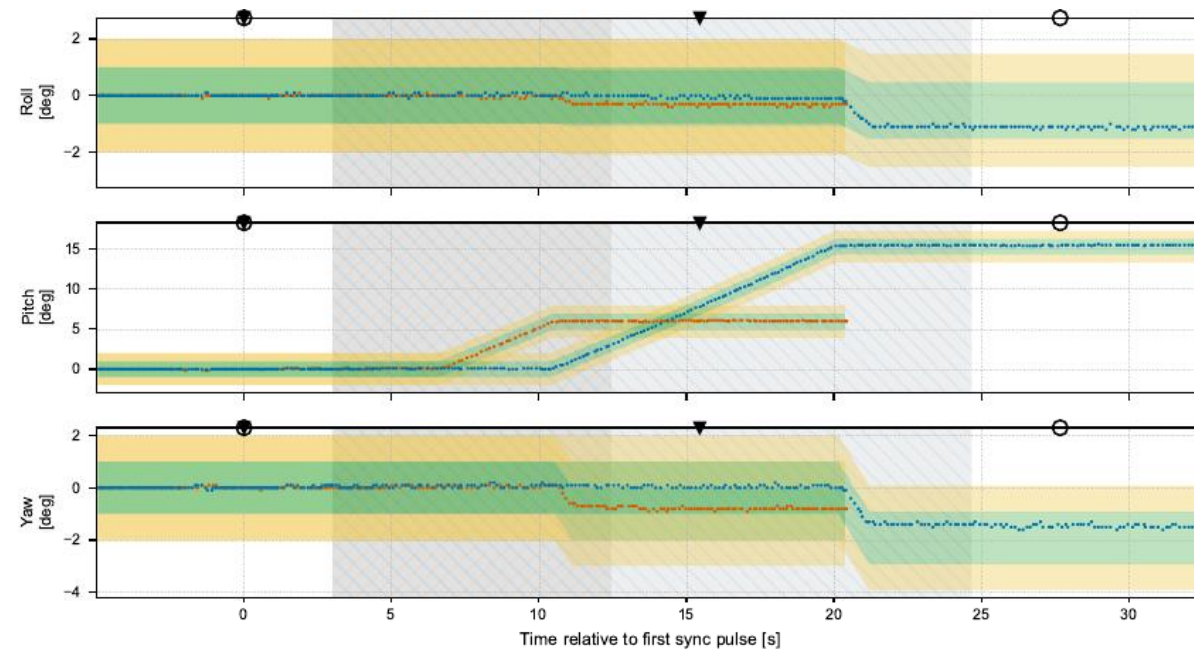
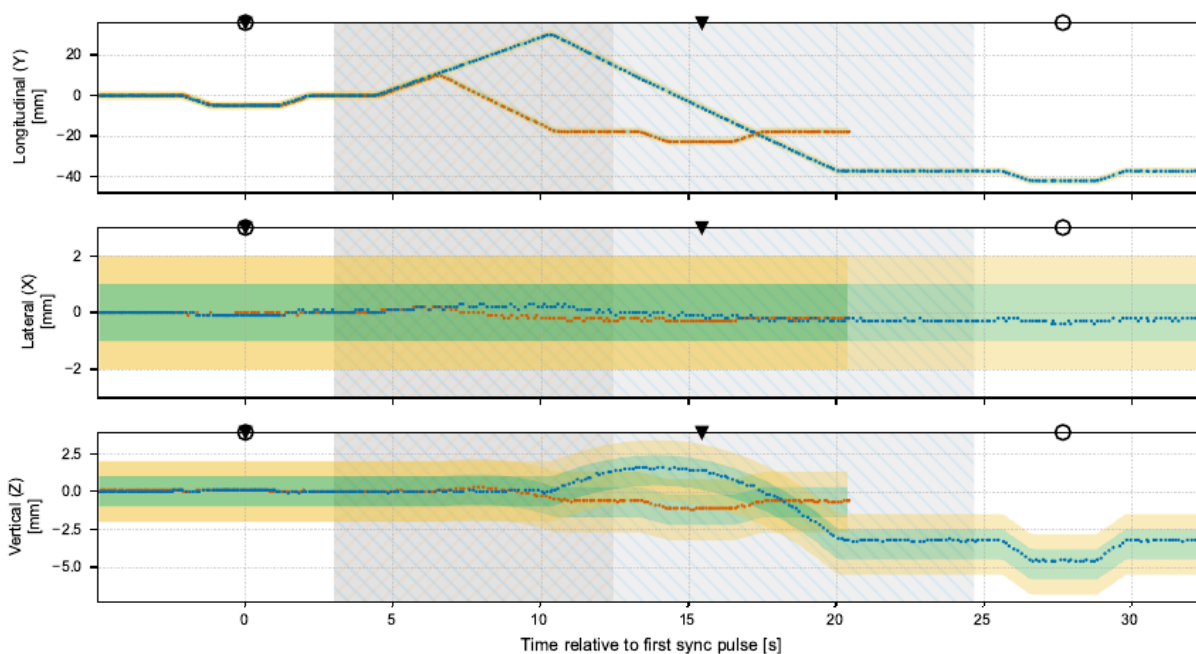
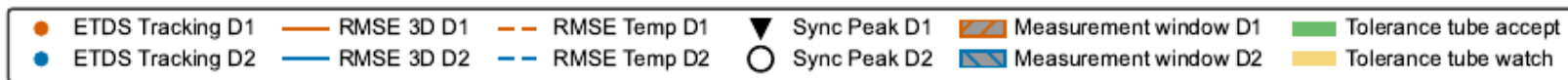


Fig. 9 Annual-QA overview page, Linac 1 at 32 °C — both deflections on one time base; grey band = evaluation window

First clinical campaign: four linacs, one verdict

4

linacs · annual reports issued
(2026)

~570

samples per DoF and linac in the
evaluation window

100 %

within 1 mm / 1° — no watch, no
action points

Annual QA 2026 — all four ETD-equipped linacs, all six DoF within tolerance

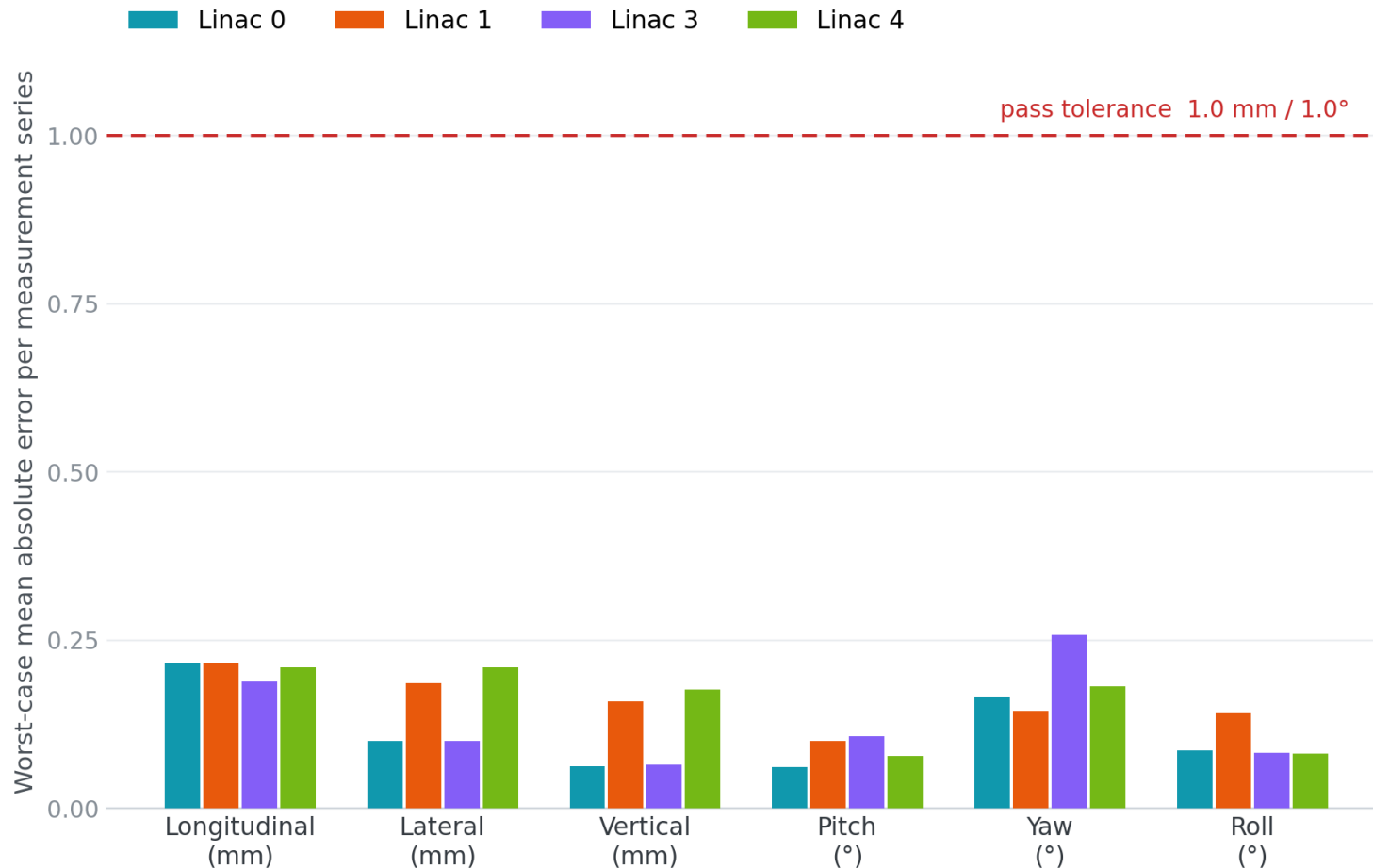


Fig. 10 Annual QA 2026: worst-case MAE per DoF and linac

Built to be copied

- Independent QA needs independent tools
- Only a time-resolved 6-DoF needed → vendor-independent
- Open repos



hardware · firmware · pipeline · SOP

ag-kollotzek.github.io/novalis-circle-2026



Take home

A guideline became a number

ESTRO-ACROP's "1 mm / 1°" is now an auditable, reproducible annual measurement.

It is clinically feasible

Half hour per linac, scripted from raw data to a signed report — run on four linacs in 2026, all green.

It is yours to copy.

Open hardware, open code, open SOP — a basis for benchmarking SGRT across institutions.



Thank you — T. Buck · N. Bersenkowitsch · L. Bodner · P. Entfellner · U. Ganswindt · Medical University of Innsbruck / Tirol Kliniken

What a phantom cannot test: clinical acceptance

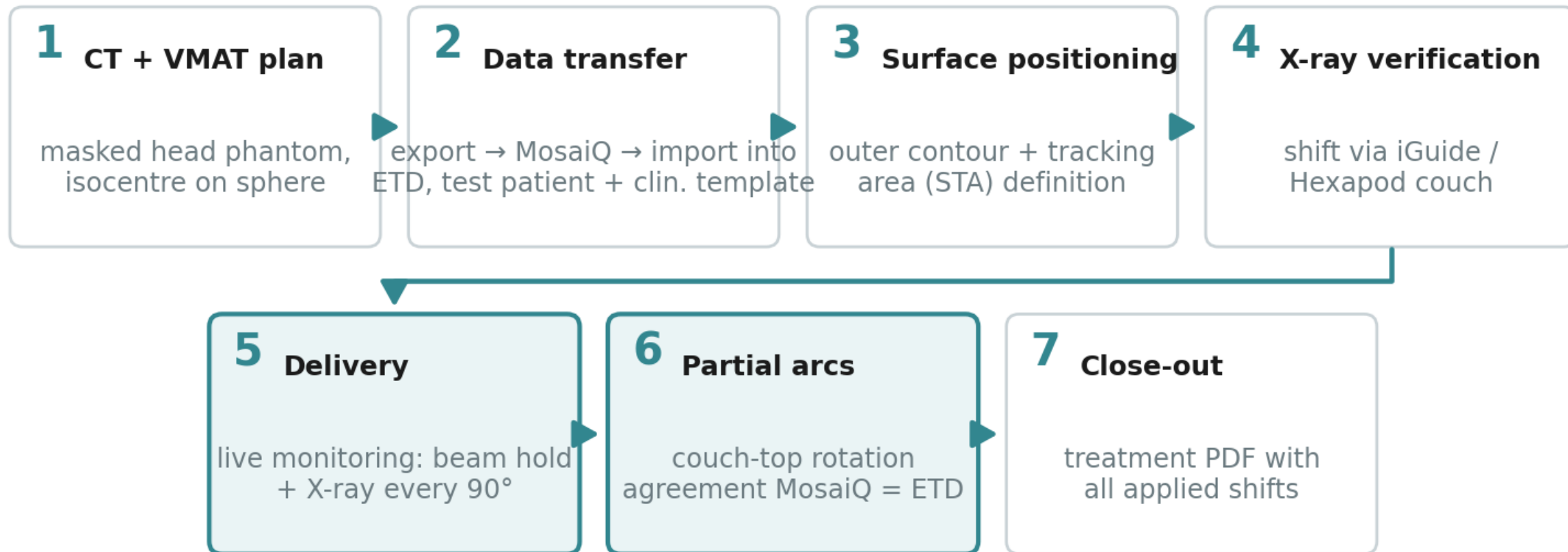


Fig. 3 Functional acceptance workflow — the full clinical path, end to end (SRS case)

Backup — Sphere eccentricity & kinematic calibration

- X-ray at extreme rotations revealed: sphere sits 1.1 mm off the rotation axis
- Apparent shift up to 1.5 mm at 90° — fully described by $2r \cdot \sin(\theta/2)$
- Corrected by a per-linac radius calibration of the kinematic model
- RMS deviation: 0.52–0.74 mm (as-built) → 0.10–0.27 mm (calibrated)
- A ground truth that detects its own sub-millimetre build errors is working as intended

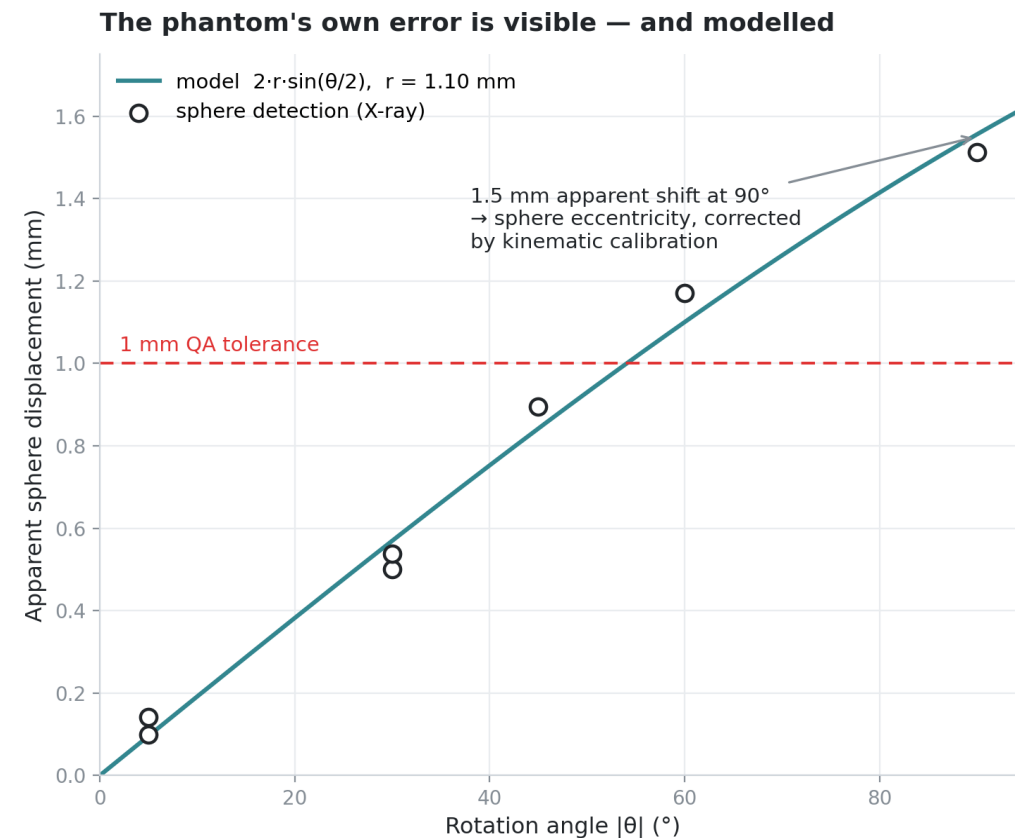


Fig. B1 Rotation series — sphere detection vs. eccentricity model

The phantom's own error is visible — and modelled: proof that the X-ray arbiter is sensitive.

Next entries in Table 4

- D3 respiratory trace: removable head → breathing breast phantom
- Gated vs. ungated on EPID — from tracking accuracy to delivered dose
- P1 thermal drift · A5 couch rotation (data acquired) · E1 end-to-end
- Open: independent CT verification & multi-point linearity

Next entries in ESTRO-ACROP Table 4

D3

Respiratory trace

removable head → breathing breast phantom

P1

Thermal drift

A5

Couch rotation

measurement series already acquired

E1

End-to-end positioning

The platform grows along the guideline: breathing, thermal drift, couch rotation, dose.